

DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Program: B.Tech. Semester-IV (ECE and AI & ML)
Course Title: Applied Linear Algebra
Course Code: MAIC 204

Paper: REGULAR and REAPPEAR
Session: 2024-25
M.M. 50

Note: Attempt all the Questions.

1. Let W_1 and W_2 be subspaces of a vector space V such that the set theoretic union of W_1 and W_2 is also a subspace. Prove that one of the spaces W_i is contained in the other.

2. If W is a subspace of a finite dimensional vector space V , then show that every linearly independent subset of W is finite and is part of a (finite) basis for W .

3. Let V be the vector space of all $n \times n$ matrices over the field F , and let B be a fixed $n \times n$ matrix. If $T(A) = AB - BA$

verify that T is a linear transformation from V into V .

4. Let V be the space of $n \times 1$ matrices over F and let W be the space of $m \times 1$ matrices over F . Let A be a fixed $m \times n$ matrix over F and let T be the linear transformation from V into W defined by $T(X) = AX$. Prove that T is the zero transformation if and only if A is the zero matrix.

5. Find two linear operators T and U on R^2 such that $TU = 0$ but $UT \neq 0$.

6. State and prove Gram-Schmidt orthogonalization process.

7. Find the characteristic polynomial and minimal polynomial of the following matrices:

(a) $\begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & -2 & 2 \\ 4 & -4 & 6 \\ 2 & -3 & 5 \end{bmatrix}$

8. Solve by LU decomposition method

$$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ 4x_1 + 3x_2 - x_3 &= 6, \\ 3x_1 + 5x_2 + 3x_3 &= 4. \end{aligned}$$

9. (a) Find all possible Jordan Canonical forms for those matrices whose characteristic polynomial $c(x)$ and minimal polynomial $m(x)$ are as follows:

$$c(x) = (x-2)^4(x-3)^2, \quad m(x) = (x-2)^2(x-3)^2.$$

(b) Determine whether the following quadratic form q is positive definite:

$$q(x, y, z) = x^2 + 2y^2 - 4xz - 4yz + 7z^2.$$

10. (a) Consider the following polynomial in $P(t)$ with the inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$:

$$f(t) = t + 2, \quad g(t) = 3t - 2, \quad h(t) = t^2 - 2t - 3$$

(a) Find $\langle f, g \rangle$ and $\langle f, h \rangle$.

(b) Find $\|f\|$ and $\|g\|$.

(b) Consider the mapping $F: R^2 \rightarrow R^2$ defined by $F(x, y) = (3y, 2x)$. Let S be the unit circle in R^2 , that is, the solution set of $x^2 + y^2 = 1$. Then describe $F(S)$.